

Sumit Jha

Software Engineer

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Experience

PiHex Labs / PythonAISolutions

Software Engineer

Dec. 2024 – Present

Remote

- Built an AI-driven medical imaging web application ("Agentic Cervical Screener") for real-time cervical cytology analysis, combining a FastAPI backend and a TypeScript/Vite frontend. Integrated a YOLO-based deep learning model for classification with an interactive NiiVue WebGL viewer for slide navigation and overlaid results.
- Contributed extensively to NiiVue (open-source WebGL2 neuroimaging viewer): developed a reusable UI component kit (controls for brightness, contrast, slice navigation, toggles) and implemented a unified canvas overlay system to improve rendering performance and fix layering issues (PRs 1334, 1356).
- Architected a secure scheduling web app for a research facility with enterprise SSO via Microsoft Entra ID (Azure AD) replacing legacy LDAP authentication. Built robust CI/CD pipelines (GitHub Actions with automated testing) and containerized deployment (Docker Compose), with modular configuration management (Pydantic) for reliability and scalability.
- Implemented Infrastructure-as-Code (OpenTofu/Terraform) to manage AWS cloud infrastructure (bastion host, site-to-site VPN, Kubernetes cluster), enabling repeatable deployments and a smooth migration of multiple websites from legacy hosting.
- Developed internal tools including the corporate website (Next.js, TypeScript, Tailwind CSS, MDX with Framer Motion) and "Project Foundry" (monorepo for project specs with standardized templates), enhancing the company's web presence and streamlining project planning.
- Worked with structured datasets in Python, performing data cleaning & transformation to support analysis & visualization.

Essentia.dev

Software Developer

Dec. 2023 – Dec. 2024

New Delhi

- Contributed to building an open-source Python library (OpenBharatOCR) for OCR of government documents.
- Collaborated with a team of developers to design and implement core features of the OCR system, enhancing its accuracy and performance. Utilized technologies such as Python, TensorFlow, and OpenCV to build and train machine learning models for text recognition.
- Implemented pre-processing techniques for image correction, noise reduction, and text segmentation to improve OCR accuracy.
- Built Python-based preprocessing pipelines to clean & normalize document datasets prior to downstream analysis.

TruAct

Software Developer

Apr. 2023 – Nov. 2023

New Delhi

- Developed custom applications on the Zoho Creator low-code platform.
- Developed a Procurement-to-Purchase (P2P) application for recording & processing purchase requisitions.
- Developed various business workflow models (Approval Chain, Hiring, Farming, Inventory Management).
- Implemented multiple API integrations between Zoho Creator and external platforms.
- Developed a monitoring and training program for interns to improve performance and skills.

Open Source Contributions

- [jupyterlab-conda-store](#), [sympy](#), [jupyterlab-pioneer](#), [nebari](#), [nebari-docs](#), [Panel](#), [niivue](#), [openbharatocr](#)

Skills

Languages Python, TypeScript, JavaScript, HTML/CSS, SQL (PostgreSQL), Deluge

Backend & APIs Django, Flask, FastAPI, SQLAlchemy, Alembic, Pydantic, REST APIs, OAuth2, Microsoft Entra, LDAP

Data Libraries & Practices Pandas, NumPy, Data cleaning, data validation, exploratory analysis

Frontend React, Next.js, Tailwind CSS, Bootstrap, jQuery, DOM, WebGL, NiiVue

DevOps & Infrastructure Docker, Docker Compose, GitHub Actions, CI/CD, AWS (Familiar), GCP, Nginx (Familiar), Terraform / OpenTofu

Databases PostgreSQL, MongoDB, SQLite

Architecture & Tooling Spec, SpecKit, System Design, Monorepos, Git, Postman, npm, Yarn

Projects

Scheduler Built a secure, scalable scheduler with enterprise SSO (Azure AD) authentication, Alembic migrations (PostgreSQL), and modular config management using Pydantic.

Agentic Cervical Screener AI-driven cervical screening web app (FastAPI + YOLO model) for analyzing cytology images, with an interactive NiiVue viewer for visualization.

Entra Credentials Validator CLI and browser tool to verify OAuth2/OIDC flows with Microsoft Entra ID (Azure AD), used to test and debug enterprise SSO configurations.

Python AI Solutions Website Corporate website built from scratch using Next.js, TypeScript, Tailwind CSS, MDX, and Framer Motion, optimized for static deployment and performance.

Website Management Platform Infrastructure and deployment management system for hosting and maintaining multiple production websites using Docker, cloud storage, and CI-driven workflows.

Project Foundry Internal project specification and planning system used to standardize architecture documents, technical proposals, and implementation workflows across teams.

Gitstats Calculates the contributors in your public repositories.

openbharatocr Open-source Python library for OCR of Indian government documents.

P2P Procurement-to-Purchase (P2P) application for recording and processing internal purchase requisitions.

Calendar-App A full-stack calendar app using Django (backend) and React (frontend), enabling users to create, update, delete, and view events in monthly and weekly views.

COVID-19-Dashboard-using-Panel An interactive COVID-19 dashboard built with Panel, showcasing Python's ability to create dynamic web apps without HTML, CSS, or JavaScript.

Education

Vivekananda Institute of Professional Studies

Master of Computer Applications

2021 – 2023

New Delhi, India

Integrated Institute of Technology

Bachelor of Computer Applications

2017 – 2021

New Delhi, India

Talks & Presentations

Lightning Talk

"Panel – The Powerful Data Exploration & Web App Framework for Python"

PyCon India 2024

Bengaluru, India

Certificates

Python and Django Full Stack Web Developer Bootcamp by Udemy

The Python Programming Comprehensive Bootcamp by Udemy

JavaScript (Intermediate) by HackerRank

Problem Solving (Basic) by HackerRank

CSS by HackerRank

Enhancing Soft Skills and Personality by NPTEL